MILITARY SPECIFICATION

RADAR BEACON
AN/APN-132 (*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive interrogations from X-Band radars and transmit a reply to the interrogation on another frequency within the X-Band. The radar beacon is used to extend the operating range of the radar.

2.1 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Receiver-Transmitter	RT-(*)/APN 132	3.4.1
Control	C-(*)/APN 132	3.4.2
Mounting	MT-(*)/APN 132	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for

FSC 5841

Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

MIL-N-16307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
SAR-300	Material Changes and Material Bulletins, Preparation of

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 21.0 pounds.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:58Z
Check the source to verify that this is the current version before use.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 150 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 6000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 350 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment. The manufacturer's recommendation of servicing such as the replacement of the magnetron will be adhered to.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.5 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. See 6.4.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% RH
Input power voltage	115 ± 1.0 V, 400 ± 10 CPS

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:58Z
Check the source to verify that this is the current version before use.

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein.

3.2.11.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400.

3.2.11.2 Shock - The equipment shall operate satisfactorily when subjected to a shock of 50 G (11 millisecond duration), on any axis.

3.2.11.3 Acceleration - The equipment shall operate satisfactorily when subjected to an acceleration of 50 G, on any axis.

3.2.12 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.2.13 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V Category "B"

3.2.13.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.14 Standby Provisions - The equipment shall provide for a standby position whereby rated voltages shall be applied to all tube filaments.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Frequency Characteristics - The equipment shall operate in the "X" Band of frequencies as follows:

3.3.2.1 Receiver - The center of the receiver pass band shall be adjustable to any frequency from 8500 to 9600 MC.

3.3.2.2 Transmitter - The transmitter frequency shall be adjustable to any frequency from 9300 to 9400 MC, by the installation of one of two magnetrons. Each magnetron shall be adjustable over a frequency range of 50 MC.

3.3.2.2.1 Factory Tuning - Unless otherwise specified in the applicable contract, the equipment shall be delivered with the transmitter tuned to 9325 MC (one magnetron).

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:58Z
Check the source to verify that this is the currentversion before use.

3.3.3 Receiver Characteristics -

3.3.3.1 Interrogation Signal - The equipment shall receive radar interrogation signals. The interrogation signal may be either single or paired pulses having a pulse duration of 0.5 ± 0.1 microsecond. The paired pulses shall be adjustable, at the equipment, from 1.5 to 6 microseconds in 0.1 microsecond increments. The rise time of the interrogation signal will be 0.1 microsecond or less and the fall time 0.2 microsecond or less.

3.3.3.2 Trigger Sensitivity and Stability - The equipment shall have a trigger sensitivity of at least -65 dbm over a frequency range of ± 2 MC from the nominal frequency with less than 11 random transmitter pulses per minute. Full triggering shall be defined as at least 99% replies.

3.3.3.3 Selectivity -

3.3.3.3.1 Bandwidth - The equipment shall have a bandwidth of at least 10 MC at the 3 db points. The bandwidth may be increased to allow for frequency stability of the local oscillator and i-f amplifier to assure performance as specified in 3.3.3.2.

3.3.3.3.2 Image - The image frequency shall be at least 100 MC from the nominal frequency. No preselection shall be used within the Receiver-Transmitter unit.

3.3.4 Decoder Characteristics -

3.3.4.1 Single Pulse Interrogation - The equipment shall respond to a single pulse interrogation having the following characteristics:

(1)	Pulse width	0.5 ± 0.1 microsecond
(2)	Pulse rise time	0.1 microsecond or less
(3)	Pulse fall time	0.2 microsecond or less

3.3.4.2 Double Pulse Interrogation - The equipment shall respond to a double pulse interrogation having the following characteristics:

(1)	Pulse space	1.5 to 6 microseconds in 0.1 microsecond increments
(2)	Pulse width	0.5 ± 0.1 microsecond
(3)	Pulse rise time	0.1 microsecond or less
(4)	Pulse fall time	0.2 microsecond or less

3.3.4.2.1 Double Pulse Reject Limit - The equipment shall reject interrogations having a pulse spacing of less than 1.5 microseconds or greater than 6 microseconds when in the double pulse mode of operation.

3.3.4.3 Interrogation Control - The mode of interrogation to which the beacon will respond to shall be remotely controlled.

3.3.5 Transmitter Characteristics -

3.3.5.1 Frequency Stability - After a 5 minute warm up, the transmitter frequency shall be within ± 2 mc of the nominal frequency under standard conditions. After a 15 minute warm up, the transmitter frequency shall be within ± 5 mc of the nominal frequency under extreme service conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the current version before use.

3.3.5.2 Reply Signal Characteristics - The equipment shall reply (to properly coded interrogations) with a single pulse having the following characteristics:

(1)	Pulse width	0.5 ± .1 microsecond
(2)	Pulse rise time	0.1 microsecond or less
(3)	Pulse fall time	0.2 microsecond or less

3.3.5.3 Transmitter Power - The peak pulse power from the transmitter antenna terminal shall be at least 100 watts at duty cycles as high as 0.5 percent.

3.3.5.4 Overload Protection - Circuitry shall be provided to limit the maximum duty cycle of the transmitter to 3 per cent. This circuitry may utilize the recovery time of the beacon provided there is no degradation in performance from that specified in 3.3.8.

3.3.5.5 Transmitter Oscillator Load Isolator - The circuitry shall include a load isolator in the r-f output from the transmitter to prevent "long line" affects of the antenna and transmission line from detuning the transmitter an excessive amount. The load isolator shall provide an isolation factor of at least 15 db.

3.3.6 Beacon Delay Time - The delay time between the single interrogation pulse or the second pulse of the double pulse interrogation and the beacon reply pulse shall be 1.0 ± 0.1 microsecond.

3.3.7 Random Triggering - The beacon shall not trigger itself due to random noise at a rate greater than 10 per minute, averaged over a 10 minute interval, when the receiver r-f input is terminated with a dummy load.

3.3.8 Beacon Recovery Time - The beacon recovery time shall not exceed 20 microseconds and the receiver sensitivity shall have returned to at least -62 dbm 20 microseconds after the receipt of an acceptable interrogation signal.

3.3.9 Antenna Terminal - The receiver and transmitter shall have separate antenna connections.

3.3.10 Maximum Signal Input - The maximum signal input for operation within the specification limits will not exceed 0 dbm.

3.3.11 Receiver Protection - The transmitting antenna will be separated from the receiving antenna such that at least 40 db of isolation will occur or a suitable duplexer will be used when a common antenna is employed.

3.4 Detail Requirements -

3.4.1 Receiver-Transmitter RT-(*)/APN 132 - The receiver-transmitter unit shall meet the following requirements.

3.4.1.1 Function - The receiver-transmitter receives interrogations from X-Band radars operating on the same frequency that the beacon receiver is tuned to; the interrogation signal is decoded and is used to trigger the beacon transmitter that is tuned to another frequency within the X-Band.

3.4.1.2 Form Factor - The unit or units, excluding mounting and connectors shall be the minimum practicable, and shall not exceed 14.5 inches long.

3.4.1.3 Weight - The weight of the receiver-transmitter shall not exceed 21 pounds.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the current version before use.

3.4.1.4 Contents of Unit - The receiver-transmitter unit shall comprise subassemblies that perform the following functions:

(1) Receiver local oscillator

(2) Receiver mixer

(3) Receiver IF amplifier

(4) Video amplifier

(5) Decoder

(6) Modulator

(7) Magnetron

(8) Load isolator

(9) Power supply

3.4.1.5 Controls - The following controls shall be provided on the external surface of the receiver-transmitter.

(1) Magnetron frequency control

(2) Receiver local oscillator (klystron) frequency control

(3) Klystron reflector voltage control

(4) Trigger level control

(5) Paired Pulse adjustment

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided by standard receptacles in accordance with MIL-E-5400 as follows:

Reference Designation	Receptacle Type	Function
J1		Power inlet and control panel outlet
J2		Transmitter Antenna
J3		Receiver Antenna

3.4.2 Control C-(*)/APN 132 - The control shall meet the applicable requirements of Specification MIL-C-6781 for Type I control panel.

3.4.2.1 Function - This unit shall provide controls for turning on the equipment and selecting the type of operation.

3.4.2.2 Form Factor - This unit shall have a maximum allowable size of 5.75 inches wide by 1.5 inches long by 2 inches deep.

3.4.2.3 Weight - The weight of this unit shall not exceed 0.75 pound.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:58Z
Check the source to verify that this is the current version before use.

3.4.2.4 Controls - The following controls shall be provided:

(1) "PWR" (Power) - A two-position toggle switch shall be provided for controlling the power to the equipment. In the down position labeled "OFF" no primary power shall be furnished to the equipment. In the up position labeled "PWR," the equipment shall be energized.

(2) "Operate" - a two-position toggle switch shall be provided for controlling the operation of the equipment. In the down position labeled "STDBY" no replies shall be made to interrogations. In the up position labeled "ON" the beacon shall be fully operative.

(3) "Mode" - A two-position toggle switch shall be provided for controlling the mode of operation of the equipment. In the down position labeled "SINGLE" the beacon shall respond to single pulse interrogations. In the up position labeled "DOUBLE" the beacon shall respond to double pulse interrogations.

3.4.2.5 Electrical Connections - Connections to external circuits shall be provided by means of a standard connector in accordance with Specification MIL-E-3400.

3.4.3 Mounting MT-(*)/APN-132 - This mounting shall be a vibration isolator type for mounting the receiver-transmitter and shall meet the applicable requirements of Specification MIL-C-172.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T12:58Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment. It may not be resubmitted as the production sample model.

Source: https://assist.dla.mil -- Downloaded: 20261019T12:58Z
Check the source to verify that this is the current version before use.

4.3 Production Sample Tests - One early production equipment shall be selected and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. Final acceptance of this equipment shall be after completion of the production sample tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when required by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - After completion of the production sample tests the equipment shall be returned to the contractor for any necessary rework. The equipment shall then be resubmitted for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the currentversion before use.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the currentversion before use.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the currentversion before use.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:58Z
Check the source to verify that this is the currentversion before use.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:58Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use - The radar beacon shall be for use in naval aircraft in conjunction with radar direction finding and ranging to increase the range capability of the system.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 Definitions - The following definitions shall apply:

6.6.1 Interrogation - An interrogation is one or two radio frequency pulses transmitted by an interrogator or radar set.

6.6.2 Reply - A reply is a radio frequency pulse transmitted by a beacon as a result of an interrogation.

6.6.3 Pulse Width - Pulse width is that interval of time between points at which amplitude of the pulse is 50% of the pulse amplitude.

6.6.4 Pulse Amplitude - Pulse amplitude is the maximum voltage amplitude of the pulse or r-f pulse envelope, excluding spike.

6.6.5 Pulse Rise Time - Pulse rise time is that interval of time required for the leading edge of the pulse to rise from 10 to 90% of the pulse amplitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:58Z
Check the source to verify that this is the current version before use.

6.6.6 Pulse Fall Time - Pulse fall time is that interval of time required for the trailing edge of the pulse to decay from 90 to 10% of the pulse amplitude.

6.6.7 Pulse Spacing - Pulse spacing is that interval of time between 50% pulse amplitude points on the leading edges of the pulses involved.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:58Z
Check the source to verify that this is the current version before use.